Chapter 21
A Non-classical Acoustics Teaching Lab Supported by BYOD and Inquiry-Based Learning



Antonella Longo, Marco Zappatore, and Marisa Michelini

Introduction

The scientific educational scenario is nowadays populated by several apparently appealing opportunities disclosed by novel technologies. Their presence alongside traditional educational activities, however, must be careful evaluated in order to avoid that the improvements in didactic and learning qualities, typically mentioned as one of the core benefits brought by technological advancements, would represent neither a pedagogical risk nor a false myth.

A series of key elements must be considered:

1. The availability of portable devices with considerable technological capabilities amongst students and teachers;
2. More broadly, the daily technological scenario where students and teachers live when they are outside schools;
3. The technological offering within didactic coursework.

As for the first aspect, it has been already assessed by several studies how mobile devices are incredibly diffused amongst students, especially younger ones. Since the early 2010s, mobile devices have become students' inseparable friends for their non-educational activities so that a "technology addiction" can be clearly highlighted amongst them (Raman and Pramod 2014).

The diffusion of mobile devices in young users was already well known, due to the motivational factors supporting this scenario, ranging from dependent and

A. Longo · M. Zappatore (✉)
Department of Innovation Engineering, University of Salento, Salento, Italy
e-mail: antonella.longo@unisalento.it; marcosalvatore.zappatore@unisalento.it

M. Michelini
Physics Education Research Unit, University of Udine, Udine, Italy
e-mail: marisa.michelini@uniud.it

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compulsory behaviours (Hooper and Zhou 2007) to emotional and instrumental ones (Ishii 2011). However, nowadays the diffusion is so spread that "*mobile devices are not just landline and computer substitutes but have become personal extension for the Millennial Generation students*" (Ahmed 2012).

As a direct consequence, a considerable amount of educational activities has started to embed portable electronic devices as possible didactic supports in the recent years, under the broad definition of mobile learning (Pachler et al. 2009; Traxler 2010).

However, despite numerous efforts, relevant gaps still exist between technology-intensive activities that students perform outside their schools and the level of technology available within the educational offering they receive daily (Wu et al. 2012). This has been recently confirmed by the Pearson Student Mobile Device Survey (Harris Poll Interactive Online Inc. 2015), according to which, students:

1. Would like to use mobiles more often in the classroom (Elementary Schools, ES: 72%; Middle Schools, MS: 66%; High Schools, HS: 54%)
2. Show a predominant usage of smartphones (ES: 53%; MS: 66%; HS: 82%)
3. Also show a high and growing usage of tablets (ES: 78%; MS: 69%; HS: 49%)
4. Believe in the didactic effectiveness of mobiles (overall: 90%)
5. Believe mobiles make learning more fun (overall: 89%)
6. Believe mobiles are more suitable to their learning behaviours (overall: 82%)
7. Believe mobiles can improve their learning outcomes (overall: 81%)
8. Believe they know mobiles better than their teachers (ES: 62%, MS: 72%; HS: 77%)
9. Attend few schools where mobiles are provided for every student (overall: 19%).

But how this positive attitude of students towards mobile devices, intended as profitable supports along the educational path, and how this evident limitation of schools in providing an adequate number of technologically-advanced portable devices can be actually tackled? And, at the same time, how mobile devices have actually proven to be effective in this field?

Several educational paradigms have been proposed and already analysed so far, ascertaining an undoubted usefulness for mobile devices in education, provided that the corresponding risks are clearly highlighted and coped with accordingly. In this research work, we have referred to two of these paradigms. On the one hand, the Bring Your Own Device (BYOD) paradigm is based upon the conception students can introduce their personal electronic devices into their learning experiences with a significant increase in learning effectiveness (Afreen 2014). This is particularly evident in STEM (Science, Technology, Engineering and Math) disciplines, where modern mobile devices offer powerful computational and networking capabilities. On the other hand, the Mobile Crowd Sensing (MCS) paradigm allows leveraging mobiles also as effective sensing stations, thanks to their rich set of built-in sensors (Ganti 2011), thus making them suitable for laboratory-oriented educational activities.

We believe that a proper combination of the BYOD and MCS paradigms can help overcoming current financial and logistic limitations in laboratorial didactics, which

are normally due to high buying and maintenance costs for equipment as well as to the need of skilled personnel for lab assistance activities. Moreover, we also believe in mobiles as suitable technological drivers for engaging students more effectively in Inquiry-Based Learning (IBL) activities (Abd-El-Khalick et al. 2004), which promote the active participation of learners in problem solving.

Amongst the possible Laboratorial Didactic Experiences (LDEs) that can benefit from the adoption of MCS and BYOD, we have selected the acoustic domain for several reasons. First, the study of acoustic and sound propagation mechanisms usually falls outside the scope of the majority of scientific laboratories in middle and high schools. Second, the constantly increasing attention of the public opinion towards noise pollution, especially in urban scenarios, represent a significant additional element of interest. By involving students in studying these topics, a better awareness of the noise-related health issues can be achieved as well.

Therefore, we describe in this paper our IBL-based LDE in acoustics, which addresses students from middle and high schools into IBL-based LDEs and exploits MCS and BYOD paradigms. A technological platform has been developed for allowing students to participate in sound monitoring campaigns and acoustic-oriented lab activities.

The paper is organized as follows: Section "Adopted Technological and Educational Paradigms, Strategies and Approaches" summarizes the educational relevance of the BYOD paradigm and IBL strategies. Section "The Platform" presents the developed platform. Section "The Laboratorial Didactic Experience (LDE)" provides a detailed overview of the proposed LDE. Section "Validation Methodology and Discussion on LDE Outcomes" discusses the didactic outcomes. Conclusions are reported in the last Section.

Adopted Technological and Educational Paradigms, Strategies and Approaches

The educational effectiveness of the BYOD paradigm has been largely discussed in the recent years. On the one hand, several benefits have been clearly identified, such as: easier interaction across teachers and students, extended access to networked technologies, suitability for flipped learning and independent learning experiences (Traxler 2010; Stavert 2013). On the other hand, a series of challenges for the involved students have been also highlighted, such as: loss of attention, raise of social and technological gaps for lower income students, device misuse, security breaches in hosting schools (Gidda 2014; Bruder 2014).

Therefore, BYOD-mediated experiences would be improved if carried out by complying with specific educational strategies that foster the benefits coming from BYOD. In this sense, the IBL approach (Abd-El-Khalick et al. 2004) can be considered. In IBL, students construct their knowledge by formulating hypotheses and testing them by making observations and performing experiments (Pedaste et al. 2012). This means

students are involved in self-directed, inductive-deductive learning processes where they perform simple experiments to investigate how dependent and independent variables are related, thus introducing learners to true scientific discovery processes. The effectiveness of such an instructional approach (Alfieri et al. 2011) has been investigated in several research works in correlation with the adoption of new digital technologies and devices, demonstrating that mobile devices are particularly useful during the inquiry phases, from elementary schools (Song 2014; O'Banyon and Thomas 2014) to universities (Barlow and Liaison 2015) and even remote laboratorial learning (de Jong et al. 2014).

From a pedagogical perspective, one of the key aspects in IBL is represented by the definition and activation of the inquiry cycle, so that students could be effectively stimulated to achieve practical and conceptual efficiency during their learning experiences. In this paper we refer to the Prevision-Experiment-Comparison (PEC) approach for the inquiry cycle activation (Martongelli et al. 2001; Theodorakakos et al. 2010), which has been designed and thoroughly validated by the CIRD (Inter-Department Center on Didactic Research) of the University of Udine, especially in the Physics domain.

During phase one (Prevision) of the PEC cycle, students start with phenomenological observations, which stimulate them to interpret the observed event and to provide quantitative estimations about physical phenomena (e.g., "how do you think the annoyance of a given noise source could be quantified?"). In phase two (Experiment), students are engaged into experimentation, in order to perform analyses about the phenomena under observation (e.g., "observe the real-time graphical representation of the given sound source directly on your smartphone display"). Finally, in phase three (Comparison), students validate their initial assumptions and hypotheses, by comparing them with the experimental evidences so that theoretical conclusions can be derived (e.g., "enforce your assumptions by examining quantitatively the graph from the experiment and by comparing it with your initial hypotheses"). This methodology fits very effectively with the adoption of mobile-mediated learning activities.

The fruitful combination of BYOD and IBL, moreover, paves the way for a massive scaling of the laboratorial dimension, especially when it is delivered online (through both virtual and remote labs). This is mainly determined by the specific dual nature of mobiles, which can behave at the same time both as semi-professional scientific sensing station (i.e., suitable to carry out LDEs) and as informational vectors (i.e., usable by students to access and consume didactic and learning contents). In such a scenario, the so-called MOOL (Massive Open Online Labs) (Lowe 2014) are becoming more and more spread out. The authors have already started exploring the effectiveness of the BYOD approach in STEM didactics, with promising results. Indeed, both the educational (Longo et al. 2015a) and technological (Longo et al. 2015b) aspects have been evaluated during a LDE addressing the didactics of the Electromagnetism in a first-degree academic coursework. The achieved outcomes have demonstrated a relevant improvement in students' engagement.

Consequently, in order to assess more systematically and more rigorously from an educational perspective the suitability of our proposal, we have applied a similar

approach to the didactics of acoustics in this research work. A set of inquiry phases tailored to the acoustic domain has been designed in collaboration with the teachers from the involved middle and high schools in order to obtain a shared and agreed LDE. The LDE will be thoroughly described in section "The Platform".

The Platform

In order to offer students and teachers an LDE in acoustics and noise monitoring, a proper technological platform is needed. This solution has to exhibit both pedagogical and laboratorial peculiarities, since it must allow students performing lab activities and teachers improving the quality and effectiveness of their teachings on the same topics. Four platform component have been defined after the requirement elicitation phase:

1. *MCS-based sensing module*: it consists of a free mobile app for Android-based systems that allows students to monitor the acoustic environment they live within and to receive learning contents on noise monitoring and noise pollution topics. The mobile app emulates a semi-professional Sound Level Meter (SLM) since it allows gathering data according to regulation-compliant noise exposure quantifiers. Some of the mobile app functionalities are depicted in Fig. 21.1.
2. *Cloud-based data management module*: it is tasked to sensor data management and learning material organization.
3. *Visualization module*: it is represented by a Web-app allowing both students and teachers to visualize on interactive geo-referenced online maps the results of the acoustic monitoring campaigns carried out by their peers.

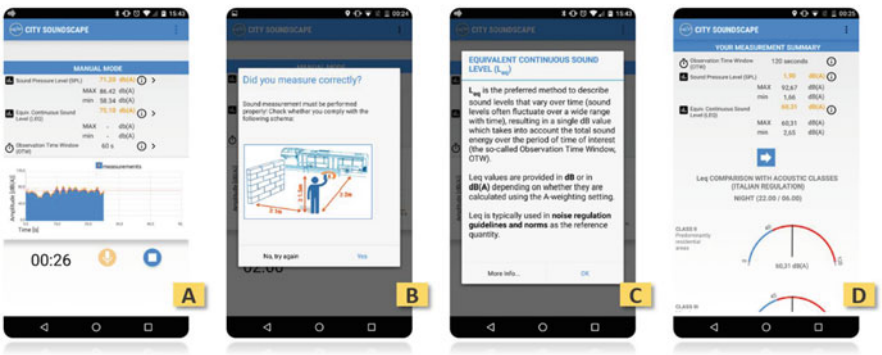


Fig. 21.1 Mobile app pages: real-time measurement (**A**); suggstions on measurement techniques (**B**); didactic material on adopted noise exposure quantifiers (**C**); measurement comparison with regulatory thresholds (**D**)

4. *Learning Management System (LMS)*: it helps managing didactic materials. It is a customized-version of the widely-adopted Moodle framework (see section "The Laboratorial Didactic Experience (LDE)")

As the technical and functional description of the proposed platform is out of scope in this context, we refer the interested readers to (Zappatore et al. 2016).

The Laboratorial Didactic Experience (LDE)

Overall Context

The LDE has involved more than 300 students and 20 teachers from 20 classes belonging to 7 different schools in Brindisi (Italy). The experimentation started at the same time in all the selected schools and it has lasted for 3 months. The experimentation has consisted of three sequential phases, according to the PEC-activated IBL approach (see section "Adopted Technological and Educational Paradigms, Strategies and Approaches"). The experimentation has consisted of three sequential phases:

1. *Preparation*: teachers from selected schools have been trained on how to introduce students to the platform and to its featuring aspects.
2. *Didactic modules*: students have been provided with learning contents from the platform. Students have also performed collaborative measurement sessions on environmental noise. The acquired knowledge has been evaluated through specific assessment questionnaires (via the LMS module) both during the modules and at their completion.
3. *Learning effectiveness analysis*: learning outcomes have been evaluated by examining the submitted questionnaires and through conclusive meetings held with the teachers.

Objectives

Primary learning objectives are geared to:

1. Engaging students in learning the core concepts in acoustics both in a descriptive and in an interpretative way.
2. Improving the laboratorial didactic experiences for students

A set of cross-disciplinary, secondary educational goals have been defined as well:

1. Improving interactions amongst teachers and students.
2. Improving interest rates in students.

3. Improving students' skills in the English language, since the majority of the accessible platform components as well as the majority of the didactic contents have been provided in English.
4. Improving students' knowledge in new technologies.
5. Fostering students' awareness in sustainable communities and environmental sustainability (since the acoustic monitoring domain is strictly related to the environmental pollution issues).

Moreover, a set of additional goals has been also defined. It enlists:

1. Disseminating a STEM-based, laboratory-oriented culture across schools.
2. Improving the overall didactic quality in STEM.
3. Fostering a positive impact on local schools (by enlarging their technological didactic offering and by motivating teachers in participating to novel IT-oriented activities).
4. Achieving tighter relationships between local schools and the local university context (since the platform has been developed by academic personnel who attended and interacted with students and teachers during the meetings).

Learning Contents

Five Learning Sections (LSs) on acoustics have been identified. For each of them, several Learning Experiences (LEs) can be proposed, each having a set of Learning Objectives (LOs) and involving a set of experiments (EXPs). The proposed LSs are: LS_1, sound sources; LS_2, sound propagation; LS_3, sound detectors; LS_4, sound pressure; LS_5, sound waves.

Let us consider, for instance, the LS addressing the topic on sound waves: its LE is "*identify how sounds propagate*", the corresponding LO is "*understand that sounds propagate as mechanical waves*" and one of the possible EXP to enforce the LO is the usage of the Kundt's tube (Sakamoto et al. 2006).

Moreover, the LDE is supported by a list of freely-available mobile apps that are presented and described in a dedicated manual, delivered in electronic format to students and teachers via the LMS module. These apps are used as didactic support during the different PDE phases, in order to allow students directly practicing the experiments for each LS on their own smartphones. The manual describes the main functionalities of each app and provides the links for its download, as well as the installation instructions. Mobile apps have been categorized into three different groups:

1. *Acoustic signal monitoring*: the apps belonging to this group allow their users monitoring sound according to several physical quantifiers and with different visualization techniques. For instance, the app developed by the authors mentioned in section "The Platform" and depicted in Fig. 21.1 allows monitoring sound w.r.t. (with respect to) two noise exposure quantifiers usually adopted in noise management regulations (i.e., instantaneous sound pressure level and time-averaged

sound level). In addition, other mobile apps from this group offer time and/or frequency analysis of the sounds acquired by the mobile embedded microphone, as they emulate sound and spectrum analysers, thus introducing students to experience the main functionalities of those equipment. The apps from this group are particularly suited for LS_1, LS_2 and LS_3.
2. *Acoustic signal generation*: these apps allow using the mobile devices for generating sound signals. They behave as frequency sound generators, therefore users can tune properly the generated signal by adjusting its waveform, volume, modulation, frequency. Teachers and students are suggested to use these apps during all the described LSs.
3. *Device sensor analysis*: these apps are not specifically related to the proposed LDE, since they allow examining the entire set of built-in sensors available in a mobile device. These apps, usually known as "*sensor boxes*", present users with a dashboard enlisting all the active sensors in their device. By selecting a given sensor, an instantaneous reading is provided, as well as a concise explanation of both the referred physical quantity and the corresponding unit of measurement. These apps are useful for introducing students to consider their own smartphone also capable of "sensing the environment", so that teachers are suggested to use the apps in the first part of the LDE.

It is important to point out that the apps are most of all suitable for a combined usage: indeed, they can be exploited during the same experiment. Let us consider an experiment belonging to the LS dedicated to the analysis of sound propagation. Students can be divided into two groups: the components of the first group use the sound generating mobile apps for creating several waveforms, whilst the students from the second group use the sound monitoring apps for observing those signals according to different perspectives (e.g., in time, in frequency, in spectrograms).

Validation Methodology and Discussion on LDE Outcomes

The LDE has involved students and teachers from 20 classes belonging to 7 different schools located in Brindisi, a city with nearly 90k inhabitants in Southern Italy. The experimentation started at the same time in all the Classes have been involved pairwise. For each couple of classes, students from the first class use the proposed platform during the LDE, while students from the second class follow a traditional teaching approach, based upon frontal lessons and without any experiential activity.

During the LDE, the following aspects have been evaluated:

1. Progressive acquisition of scientific topics on acoustic and sound,
2. Progressive acquisition of a sectorial language,
3. Progressive acquisition of usage experience for the provided app,
4. Level of motivation and personal engagement in educational activities.

Students have been surveyed and evaluated with two questionnaire types: the first one (i.e., input questionnaire) has been proposed to them before starting the LDE, in order to assess their initial background on general scientific topics and on specific topics in acoustics. The second one (i.e., output questionnaire) has been proposed after the LDE, in order to ascertain the acquired knowledge.

Therefore, at the end of the experimentation it has been possible to compare the learning outcomes from the two different approaches, since both students from the experimental and students from the traditional approach have been addressed with the same conclusive questionnaire. A specific assessment grid has been defined, based on a set of evaluation criteria assessing the following student's capabilities:

1. Observation of physical phenomena,
2. Formulation of explanatory hypotheses about the observed physical phenomena,
3. Problem formalization,
4. Application of mathematical knowledge on problem formalization,
5. Interpretation of the available datasets,
6. Explanation of the attended experiments,
7. Explanation of the adopted solving methodology.

Four overall evaluation levels (from *L1* to *L4*) have been defined, depending on whether the students show respectively *not sufficient*, *partial*, *good* or *optimal* results w.r.t. the evaluation criteria mentioned so far.

Figure 21.2 presents a chart summarizing the comparison results. The charts describe the distribution of the evaluation levels across the four examined categories of questionnaires:

1. *QIN_t*: it represents the input questionnaire performed in classes that followed the traditional didactic approach;

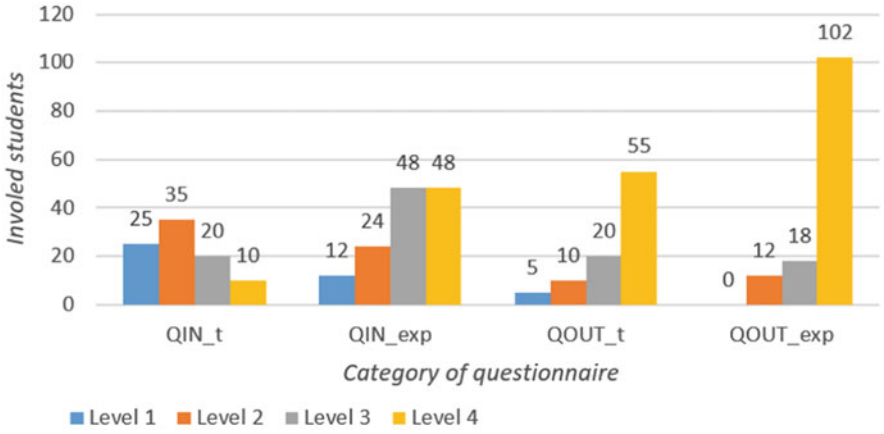


Fig. 21.2 Comparison of students' evaluation levels by questionnaire type

2. *QIN_exp*: it is the input questionnaire for classes following the experimental approach;
3. *QOUT_t*: it is the output questionnaire for the same classes from group 1;
4. *QOUT_exp*: it is the output questionnaire for the same classes belonging to group 2.

The chart reports the number of involved students in each questionnaire type. It is worth to point out that even if the validation has followed a pairwise approach, due to the inhomogeneous distribution of students across the involved classes, the participants to the LDE with the traditional didactic approach were fewer than their counterparts in the experimental approach.

As it can be easily seen, Fig. 21.2 reports a significant increase in the quality of evaluations for students, involved in experimental activities. More specifically, let us consider the proportion between negative (i.e., L1 and L2) and positive evaluations (i.e, L3 and L4) between *QIN* and *QOUT*. In the traditional approach, there are 67% of negative evaluations and 33% of positive evaluations in the *QIN* questionnaires. These percentages become 16% and 84% respectively in *QOUT*. Similarly, in the experimental approach, there are 27% of negative evaluations and 78% of positive evaluations in *QIN*. These percentages become 9% and 91% respectively in *QOUT*.

Moreover, it can be noticed that even the input questionnaire has produced better results in the classes selected for the experimentation. More specifically, the proportion between L1 and L4 in *QIN_t* is 27–11%, while the same proportion in *QIN_exp* is 9% to 36%. This is mainly determined by a better initial motivation in students from the experimental approach, who have exhibited higher interest rates from the beginning of their experience thanks to the expectation of using of mobile devices during their coursework activities.

More generally, we have detected an increased level of participation from both teachers and students in the experimental approach. Similarly, students and teachers involved in the traditional approach, as well as those who have not participated in the LDE from the remaining classes of the involved schools, have expressed the determination to participate in a similar experimentation in the forthcoming school year.

Conclusions

The recent advances in digital technologies and devices are becoming more and more suitable to be integrated into educational activities. Several methodologies and paradigms have been proposed so far in the scientific literature in order to improve both teaching effectiveness and learning outcomes in all school levels. In this paper, we refer to the Bring Your Own Device (BYOD) and the Mobile Crowd Sensing (MCS) paradigms that respectively allow students to use their own portable electronic devices during didactic activities and to use mobiles as environmental sensing stations. These two paradigms have been considered in conjunction with the Inquiry-Based Learning

methodology for designing a Learning Didactic Experience (LDE) on acoustics and sound monitoring that has involved 20 classes from 7 middle and high schools in a city in Southern Italy. The LDE has been carefully planned and implemented by complying with the Prevision-Experiment-Comparison (PEC) cycle. The paper has also addressed all the relevant aspects of the LDE, in terms of learning objectives and contents, didactic strategy and validation methodology. The promising results of the validation phase has been also discussed.

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